

Individual Investors and Option Trading: Attention Grabbing versus Long-term Strategies *

Christof Beuselinck §
Tilburg School of Economics and Management
PO Box 90153 5000 LE Tilburg (NL)
Phone: +31 13 466 2682
C.beuselinck@uvt.nl

Dries Heyman*
Ghent University
W. Wilsonplein 5D, 9000 Ghent (BE)
Dries.Heyman@UGent.be

Maarten Pronk
Erasmus University Rotterdam
P.O. Box 1738, 3000 DR Rotterdam (NL)
Pronk@ese.eur.nl

December 21, 2010

§ Corresponding author. * Part of this paper was completed when Maarten Pronk was at Tilburg University. This paper has benefited from helpful comments from Rico Von Wyss (conference discussant) and Chao Hsien Lin (conference discussant) as well as from workshop participants at the 2010 Mid-West Finance Association (Las Vegas), the 2010 FMA Europe Annual Conference (Hamburg), seminar participants at Ghent University and Tilburg University and from insightful discussions with Frank Hodge. The authors thank the anonymous retail bank who provided us with the data for this study and Dries Heyman specifically appreciates financial support from the BOF (Bijzonder Onderzoeksfonds).

*Presenting author

Individual Investors and Option Trading: Attention Grabbing versus Long-term Strategies

Abstract

This paper analyzes trading records of online retail bank investors to examine whether attention-type events dominate return feedback strategies in explaining individual investors' stock option trading decisions. We show that although individual investors are net buyers of *common stock* on abnormally high volume days, they follow contrarian investment strategies after extreme prior day stock price performance. Moreover, we find that individual investors especially exploit *stock options* to follow contrarian investment strategies in that they initiate over twice as many bullish-type (nearly half as many bearish-type) option contracts after extremely poor (good) prior-day returns. Further, we observe variance in contrarian behavior across investor types: extremely optimistic investors pursue contrarian investment strategies more (less) pronounced around highly negative (positive) prior day returns. Finally, we show that the same extreme optimists especially overweigh short-term compared to long-term return feedback information in making individual stock option trading decisions. Combined, this study provides novel insights in the dynamics of individual investors' option trading decisions and in the distinctive roles of cognitive biases underlying this process.

Keywords: Individual Investors; Option Trading; Trait Optimism; Bull-Bear

JEL CLASSIFICATION: G11, G15, G24

1. Introduction

The seminal work by Black and Scholes (1973) has spurred a multitude of work in the pricing of options and derivatives. Despite the significance of option trading for financial markets worldwide, little is still known about the determinants and consequences of trading in this important set of derivatives. Exceptions consist of work by Lakonishok et al. (2007) and Roll et al. (2010), which provides insightful evidence on the large cross-section of option trading activity and its underlying determinants. Other work has found evidence of cognitive biases in option trading activity in that option investors *underreact* to instantaneous variance in the underlying asset but *overreact* in periods of most extreme daily variance changes (Potesman 2001).

Despite these recent academic advancements, the literature has been very silent on option trading by individual investors. Reasons for this are that in most countries, including the U.S., individuals did not trade frequently in options and, as a consequence, datasets containing sufficient individual investors' stock option transactions are therefore often unavailable (Collins and Kearns 2010).¹ The Dutch environment, however, forms an exception to this rule in that private individuals already have a long history in option trading and options are an integral part of many individual household investments (Bauer et al. 2009). One of the potential reasons for the prominence of option trading in the Netherlands may be found in the fact that the

¹ Exceptions exist however, although other datasets on private individuals option trading are often self-selected and relate to managerial (CEO) stock options exercising and/or contain firm-specific option contracts with respect to a limited number of underlying common stock (e.g., Heath et al., 1999; Bell et al., 2002; Rajgopal and Shevlin, 2002). Other work on option trading by unsophisticated investors is based on pooled observations from non-market maker customers of the Chicago Board Options Exchange (CBOE) and consequently does not allow tracing of option trading behavior to individual investor characteristics (e.g. Mahani and Potesman 2008; Potesman and Serbin 2003).

Dutch pioneered short selling, option trading, debt-equity swaps, and other speculative instruments, much as we know them today (Murray 2001). In the current paper, we employ the Dutch environment to examine common patterns of individual investors' option investment behavior by relying on a large dataset of over 138,000 individual stock option transactions originated by individual investors.

An ultimate question arises: Why do individuals trade in stock options? Classical textbooks would mention examples like leveraging and hedging as usual suspects for investing in stock options, (e.g., Hull 2008). Option trading is particularly attractive for agents with borrowing constraints and strong beliefs about financial assets' future returns (Roll et al. 2010). However, Cao and Wei (2008) document larger information asymmetries for options than for the underlying stock, implying that especially informed and professional agents would find the options market a more efficient venue for trading. The fact that private individuals who are – by nature – less well informed also decide to trade in stock and index options triggers multiple questions about their underlying investment motives and choices.

Individual investors' decisions underlying the investment in common stock have been studied already extensively. Barber and Odean (2008) document that individual investors are especially net buyers of attention grabbing stocks, i.e., stocks that are in the news, stocks with abnormal trading volumes, and with extreme one-day performance. Barber and Odean explain their findings as a result of the difficulty for individuals to choose from a myriad of stocks to *buy*, making them inclined to reside to high-attention stock. However, the same is not true for common stock *sales* because individuals

typically only hold a small number of common stock positions which they effectively can sell.²

The current study examines the dominance of this attention hypothesis for individual investors in the absence of short-sales constraints, i.e. when stock option trading is easily accessible and not legally prohibited. In particular, we study whether individual investors are biased towards initiating option trades in so-called 'glitter stocks', i.e., stocks that received attention in the recent past or rather reside to short- and long-term return feedback trading strategies. In addition, we are interested in individual investors' personal traits that drive the underlying beliefs about future returns. To this end, we explore the cross-sectional variance in attention-driven option trading across investor optimism characteristics.

The dataset we employ for our analyses contains over 138,000 stock option transactions (buy-sell) for 2,127 online clients from a large anonymous Dutch retail bank and which trade in options and/or common stock for the period 2006-2007. From this set of individual option investors, we further obtain psychometric information for 373 investors (i.e., 18%) relating to an individual optimism score measured via the LOT-R standardized psychometric test (Scheier et al. 1994). We examine the importance of short-term attention in the form of extreme prior day returns and abnormal trading volumes on the average direction (bullish versus bearish) of individual stock option transactions by employing a delta-adjusted adaptation of the buy-sell imbalance metric as in Barber and Odean (2008). In addition, we test for long-return window feedback strategies in option trade initiations as in Lakonishok et al. (2007). Finally, we combine

² Attention-grabbing then becomes important as it determines the preference set for a specific stock choice.

both short- and long-term window feedback analyses for data partitions on individuals' extreme optimism scores.

Results show that individuals are net buyers of *common stock* on abnormally high volume days, yet follow contrarian investment strategies after extreme prior day stock price performance. We further find that individuals especially exploit *stock options* to follow short-term contrarian investment strategies in that they initiate more than twice as many bullish-type (nearly half as many bearish-type) option contracts after extremely poor (good) prior-day returns. This suggests that in the absence of short-sales constraints, individual investors' bias towards attention-grabbing stock is no longer observed and that short-term return information is used as a strategic investment criterion. Further, we also show that individuals incrementally rely on medium and long-term return feedback in underpinning stock option investment decisions.

Finally, and consistent with our expectations, we document that especially extreme optimists (i.e., investors that belong to the top quartile of the LOT-R optimism score) overweigh bullish (bearish) option trades more (less) in case of extreme negative (positive) prior day returns. Moreover, extreme optimists especially overweigh positive information in their *initial* stock option trades (i.e., for the initial stock option transaction made at the investor-underlying level) and correct these overly optimistic beliefs in consecutive trades. Finally, long horizon investment strategy tests as in Lakonishok et al. (2007) reveal that the same extreme optimists – as opposed to extreme pessimists – especially *underweigh* long-term compared to short-term return feedback information in their stock option investment decision. Our results are generally consistent with the

well-established theories on cognitive biases and explain cross-sectional variation in individual option trading.

Our results are robust to a number of sensitivity checks like controls for [i] delta-weighting versus equal-weighting the bull-bear imbalance measure; [ii] option strategies at the investor-level (i.e., speculative or naked versus hedging or covered transactions); [iii] initial versus subsequent option transactions at the investor-underlying level; [iv] alternative cut-off levels for identifying extreme optimists; [v] the exclusion of the most active traders (in terms of number of trades and self-declared weekly hours spent on investment portfolio); [vi] excluding investors that have less than 75 percent of their investment capital traded via the retail bank of interest; and [vii] performing return-based tests on index option transactions. Combined, all our evidence confirms the hypotheses that individual investors use both short and long-term market feedback information in their option trading behavior and that trait optimism is an important individual characteristic that determines variation in an individual's investment preference set.

Our paper adds to the literature in several ways. First, this is one of the first papers shedding some light on determinants of option trading decisions by individual investors. Second, by focusing on stock option rather than common stock trades, we are able to examine purchase and sale transactions for individuals as *equivalent* decisions. Although not always legally restricted, option trading is observed very rarely for individual investors. The current study is one of the first exploring the richness of stock option trading by individual investors and can provide insightful evidence on the

importance of search costs compared to short selling availability for individuals' investment decisions. Third, this paper yields interesting insights concerning psychological characteristics of individual investors and how these relate to economically motivated investment choices (Puri and Robinson 2007).

The remainder of the paper is structured as follows. In Section 2, we review relevant prior literature and formulate our hypotheses. In Section 3, we describe our dataset and provide details about the research design. We then show descriptive statistics and test our hypotheses in Section 4. Afterwards, we summarize a number of robustness checks in Section 5 and in Section 6, we conclude.

2. Literature Review and Hypotheses Development

There exists a substantial literature on trading in common stock by individual investors. Odean (1999), for instance, examines trading records of investors at a large discount brokerage firm. He finds that, on average, stocks that these investors buy underperform those they sell, even before considering transaction costs. Several other papers further look into how individuals trade stocks and conclude that, on average, individual investors pursue relatively naïve trading strategies (e.g., Barber and Odean 2000, 2002; De Bondt, 1998; Odean 1998, 1999).³

Barber and Odean (2008) show that individual investors suffer from a net-buying bias for attention-grabbing stock, i.e., stock that gets in the news; stock that experiences high abnormal trading volumes; and stock that realizes extreme one-day returns. By contrast, sophisticated investors like value and momentum managers suffer much less

³ Dhar and Zu (2006), however, show the importance of controlling for underlying individual investor characteristics when testing for cognitive biases, such as the disposition bias.

from such a bias and, not surprisingly, are following return feedback investment strategies. Barber and Odean (2008) explain their results by the fact that individuals are limited in screening alternative investment choices and therefore reside to those specific investments that catch their attention. The allegation that suboptimal trading is related to limited search ability, however, has not been tested directly so far and cannot be disentangled from the alternative explanation relating to the lack of shorting possibilities for individual investors.⁴

The literature is very scarce on how options and derivatives are traded by individual investors. The reason is that datasets containing option trading were not available on a large scale until recently. Lakonishok et al. (2007) study open interest and volume information for all equity option series listed at the Chicago Board Options Exchange over the period 1990-2001. Their results are consistent with motivations for betting on and hedging for directional price changes and for volatility in the underlying stock. Roll et al. (2010) examine option-stock (O/S) trading volume proportions and find this ratio to be particularly related to option deltas, trading costs and institutional holdings. Combined with the observation of increased O/S ratios around earnings announcements, they conclude that particular traders believe they hold relevant event-specific information.

⁴ Barber and Odean (2008), for instance, observe individual investment actions in the US where stock option trading by individuals was very limited. For the large discount brokerage house dataset in Barber and Odean, being one of the individual investor samples, only 0.29% of all positions are short positions (page 786, footnote 3). More recent information sources indicate a significant increase of option trading by US individual investors (e.g., Collins and Kearns 2010), although information on individual option trades are generally not publicly available.

Apart from these cross-sectional tests on option trading activity and determinants, there is only limited evidence on individuals trading options. Noticeable exceptions are Schmitz and Weber (2007) who document long-term trading patterns in the German market for option-like derivatives and Bauer et al. (2009) who find that derivative trading exacerbates the financial performance of Dutch private investors.⁵ Other studies on private individuals option trading employ often self-selected datasets in that they relate to managerial (CEO) stock options exercising and/or contain firm-specific option contracts with respect to a limited number of underlying common stock (e.g., Heath et al., 1999; Bell et al., 2002; Rajgopal and Shevlin, 2002).

An important yet unaddressed question relates to the underlying determinants of individual investors' option trading decisions. More in particular, we are interested in whether the availability of more sophisticated trading devices like stock options for a variety of individual investors makes individuals to naïvely trade in stock options or rather exploit the unique features of stock options to leverage-invest in feedback strategies.

Individual and institutional investors reside to options to improve returns, hedge risk or speculate on market performance (Collins and Kearns 2010; Lakonishok et al. 2007). Consistent with the findings of Schmitz and Weber (2007), we conjecture that individuals employ stock options particularly based on investments strategies to go long in stock that recently fell out of grace, respectively go short in favorable stock. An

⁵ Schmitz and Weber (2007) study bank-issued warrants as a proxy for stock options. These warrants are comparable to stock options in that they securitize the right, but not the obligation, to buy (call) or sell (put) a certain amount of the underlying security for a previously specified price. However, they differ from plain vanilla options in that they are issued by banks and that it is not allowed for investor to write (i.e. short-sell) warrants.

investment style that is directly associated with this behavior reflects a contrarian investment strategy. This results into our first hypothesis (H1):

H1: In the absence of short-sales constraints, individual investors long/short stock options to leverage-invest in contrarian strategies

However, it is unlikely that every investor will trade stock options for similar reasons. Some investors may want to invest in stock options to pursue a leverage effect via long call options, while others may reside to long put options to reduce the downside risk (Hull 2008). In either case, each investor forms its opinion based upon the belief of being better informed than the average investor. These opinions, however, driving option activity can be due either to correct beliefs about future stock price movements or to cognitive biases or other “behavioral” factors (Lakonishok et al. 2007).

Recent studies have illustrated that investment choices are affected by psychometric biases such as overconfidence and trait optimism (Barber and Odean 1999, 2000; Felton et al., 2004; Puri and Robinson 2007). Barber and Odean have illustrated that overconfidence results in a self-attribution bias and makes investors to trade too often; hereby high trading costs then work out negatively for ex-post net returns. Puri and Robinson (2007) refer to overconfidence as a special case of ‘bad’ optimism which coincides with bad decisions and eventually results in lower realized utilities. ‘Good’ optimism, by contrast, refers to a potentially valuable individual psychological trait to establish favorable habits or behavior that more likely leads to desired outcomes.⁶

⁶ Friedman et al. (1995) for instance find that people that are optimistic at risk for skin cancer are more likely to report use of sunscreen; and that optimistic coronary bypass patients are more likely to eating low-fat food. Both types of behavior are more likely to lead to a positive outcome: i.e., lower risk on getting skin cancer and lower risk on suffer from new bypass operations caused by fat-rich food.

Optimism is an important psychological trait that may play a role in preference for investment risk. Therefore, optimistic investors typically will continue to pursue goals, even in the face of negative information because of the generalized belief that good things will happen to them in the future (Gibson and Sanbonmatsu 2004). Pessimistic investors, by contrast, are more likely to disengage and withdraw when faced with negative information because of their generalized belief that good things will not happen to them in the future.⁷

There exists, however, little empirical evidence on investment choices of optimists versus pessimists. Noticeable exceptions come from psychology research in experimental settings. Felton et al. (2004) show that the well-documented gender difference for investment risk preferences is largely attributable to differential optimism biases in man versus female treatment groups. Gibson and Sanbonmatsu (2004) document that optimists, more than pessimists, maintain positive expectations and continue gambling after experiencing negative outcomes. Moreover, Puri and Robinson (2007) show that *extreme* optimists hold similar positive views towards the future as do mild optimists, albeit only more exaggerated.

Consistent with the view that optimists are more likely to take action because they believe that good things will happen to them, and that this view is most pronounced for extreme optimists, we expect that especially extreme optimists will initiate

⁷ Scheier et al. (2001) suggest that when faced with negative feedback optimists seek information, while pessimists suppress thoughts. The kind of information gathered is also likely to be different for optimists and pessimists. Recent research suggests that optimists have an unconscious attentional bias for positive information, while pessimists have an unconscious attentional bias for negative information (Segerström, 2001). Thus, optimists are more likely to seek information when faced with adversity, and this information is more likely to be positive.

contrarian-style (momentum-style) option trades in case of extremely bad (good) prior performance. Alternatively, we expect that extreme pessimists may distrust that prior bad performance will revert in the short run and be more convinced that prior good performance shall revert shortly. Therefore, extreme pessimists may reside less to contrarian investment strategies in case of bad prior performance and more to contrarian investment strategies in case of good prior performance. This argumentation results into our second hypothesis (H2):

H2: *Extreme optimists* initiate more bullish [less bearish] stock option trades in case of extremely bad [good] prior-day performance, compared to *extreme pessimists*

3. Data and Research Design

3.1 Option Trading History in the Netherlands

Before we describe our dataset, we elaborate on the history of option trading in the Netherlands. The Dutch setting is peculiar because stock options are commonly available for individual investors; are used to a large extent by a variety individual investors to complement an average household investment portfolio; and can be traded at low cost (Bauer et al. 2009). There are several reasons for the popularity of option trading in the Netherlands. One is that option trading has a long history in the Netherlands and goes back to the Mercantilist period when the Low Countries were pioneering ground for merchant banking and sophisticated banking activities started to emerge (Murray 2001).⁸ Trading in stock options has become particularly common place

⁸ The Low Countries refers to the Netherlands and neighboring areas in the 17th Century when Spanish Emperors were reigning the region. From a geographical perspective, the Low Countries are the historical lands around the Rhine, Scheldt, and Meuse rivers, including the modern countries of Belgium, the Netherlands, Luxembourg and parts of northern France and western Germany.

since the 1987 crash and has known a stable growth ever since (van Oostvoorn and Rijnhout 2006). An additional explanation is that in the Netherlands, pension savings are to a large extent centrally administered by the employer, making that part of people's income that is not needed for liquidity and safety savings is seen as "play money" and can more easily be freed up for investments in risky asset classes like shares and options (Statman 2004).

The combination of these factors makes that many retail and investment banks (including the one where we obtained our dataset from) actively promote option trading to investors, and allow for option trading in domestic and foreign stock options.⁹ Consistent with these explanations for the popularity of option trading, Bauer et al. (2009) report that for a large Dutch online investment bank, close to 50 percent of all individual investors' trades are in derivative-type instruments.

3.2 Data

3.2.1. Retail Investor Holdings and Transactions

The dataset is provided by a large Dutch retail bank and contains monthly investment holdings as well as individual transactions of 5,465 online clients. The retail bank labels this set of clients as "the more active and experienced individual investors that invest in equity and derivatives primarily via an online trading system". Also, clients that subscribe to this trading platform have a clear demand for relevant online

⁹ In a standard procedure, individual investors are required to sign an agreement in which the potential risks and consequences of trading in options are explicitly contracted. Banks have the privilege to accept or decline individual customers to trade in options. This procedure is similar to the procedure followed elsewhere in the world, including the US (Collins and Kearns 2010). A consequence of this is that a sample of option investors (including ours) is a self-selected one and that conclusions on individual option investors need to be interpreted bearing in mind that the individuals observed do not necessarily represent the average investor. Characteristics exposed in Table 3 confirm this in that option traders are more experienced and spend more time on their investment portfolio as compared to common stock traders.

information and can make quick investment decisions with 100% investment freedom (no MifID warnings) and invest in a variety of risky and riskless asset classes.

Holding and transaction files of these investors contain different asset classes like common stock, bonds and mutual funds, but also investments in derivative products like options for the period January 2006 - December 2007. Out of the 5,465 online investors, 2,254 investors (41%) engage in at least one option trade during the observation period and 210,835 stock option transactions occur. These numbers are in line with Bauer et al. (2009) and suggest that Dutch retail investors are well experienced in option/derivative trading.

We use two data sources to merge with our investor-level information: (1) time-series data on the different option contracts that are traded come from Euronext historical tapes; (2) data on the underlying stocks or indices of the option contracts are obtained from Datastream (a Thomson Reuters product). We merge the investor-level dataset with the daily stock option dataset from Euronext to retrieve relevant information on every single option contract. Via this selection technique we are able to match 206,534 option transactions (or 97.9% of all recorded transactions) for 2,127 investors.¹⁰

Further, option contracts generally differ by the following characteristics: call/put; underlying stock; expiration date; and strike price. Option contracts might differ by only

¹⁰ Note that of all option transactions, only 2.1% relate to foreign stock. In terms of option contracts, 95.3% (or: 17,387 out of 18,254 contracts) relate to a domestic (Dutch) underlying stock or index. The remaining contracts were predominantly written on an underlying with primary listing in the US. These percentages point at a home bias for stock options that is comparable to the one commonly observed for stock holdings (e.g., French and Poterba 1991). We excluded foreign stock options from the analyses below for reasons of consistency, but results remain qualitatively unchanged when including foreign stock option contracts.

one or more of these characteristics.¹¹ Over the two-year period 2006-2007, we find that these 2,127 investors trade 18,254 different types of option contracts. We observe a strong home-bias in stock option contracts.

3.2.2 Survey Sample Characteristics

In March 2007, all clients received an email distributed by the retail bank with an invitation to participate in a survey related to their online investment behavior. The survey was developed by the authors of this study and was entered into the trading communication system after verification with the retail bank. After accepting the invitation, participants were transferred to a secured web-based survey environment where a number of questions were asked about their investment preferences; the resources they spend to conduct investment actions; and the types of information they use. After 3 days, about 250 respondents had completed the survey. At the time when the survey was closed and withdrawn from the secured webpage (2 weeks after the initial invitation email), 985 respondents had completed the survey (18%). The average respondent survey completion time was 15 minutes.

In the online survey, we inserted various questions about investor characteristics like trading experience; time dedicated on their investment portfolio; the proportion of family wealth available for investments that is traded via the retail bank of interest; and the proportion of common stock versus closed-end funds versus derivatives in their investment portfolio. Also, we inserted a LOT-R (Lifetime Orientation Test), which is a standard psychometric test to proxy for individual optimism (Scheier et al. 1994;

¹¹ For instance, a call option on ING with maturity date August 2007 and strike price 25 EUR is treated as a different contract than a call option on ING with maturity date August 2007 and strike price 24 EUR.

Graham et al. 2010; Puri and Robinson 2007). The LOT-R has been used extensively in the psychology literature and has the advantages that it is a very credible measure; it is brief and easy to use; and has a well established interpretation. Its brevity makes it ideal for use in a survey such as ours where questionnaire length is an important consideration. In this test, respondents are asked to answer a number of questions on a 1-5 point scale ranging from "I agree a lot," "I agree a little," "I neither agree nor disagree," "I disagree a little," and "I disagree a lot." There are 6 pertinent questions and 4 filler questions. The 6 scored questions follow:

1. In uncertain times, I usually expect the best.
2. If something can go wrong for me, it will.
3. I am always optimistic about my future.
4. I hardly ever expect things to go my way.
5. I rarely count on good things happening to me.
6. Overall, I expect more good things to happen to me than bad.

Coding is done so that high values indicate optimism. Thus questions 1, 3, 6 are numerically coded with a 5 if the respondent answers "I agree a lot" and 1 if they answer "I disagree a lot." Questions, 2, 4, and 5 are coded as 5 if the respondent answers "I disagree a lot" and 1 if the respondent answers "I agree a lot." This gives us a range of mean (total) responses from 1-5 and allows for sample partitions on extreme optimists (top quartile) versus extreme pessimists (bottom quartile).

3.3 Research Design

From an empirical perspective, we test for the prominence of attention biases in option trading compared to feedback investment strategies in the short, medium and long term. For this purpose we rely on two main empirical test specifications. Both tests are explained in more detail below.

3.3.1 Short-term Attention Tests

A first set of tests relies on an adaptation of the buy-sell imbalance (BSI) measure of Barber and Odean (2008) to examine attention biases in the absence of short-sell constraints for individual investors. We create a daily imbalance measure to study trading behavior across deciles of stocks, partitioned on abnormal trading volume and previous day returns [i.e., attention groups]. In studying trading behavior of common stock around attention-events like prior day abnormal volume and extreme returns, Barber and Odean (2008) suggest a buy-sell imbalance measure expressed as follows:

$$BSI_{pt} = \frac{\sum_{i=1}^{n_{pt}} N.B_{it} - \sum_{i=1}^{n_{pt}} N.S_{it}}{\sum_{i=1}^{n_{pt}} N.B_{it} + \sum_{i=1}^{n_{pt}} N.S_{it}} \quad (1)$$

Where n_{pt} is the number of stocks in partition p on day t , $N.B_{it}$ is the number of purchases of stock i on day t and $N.S_{it}$ is the number of sales of stock i on day t . The time-series mean of the daily buy-sell imbalances can then be calculated for the days trading is available for the investors. In a first set of tests, we employ this buy-sell imbalance (BSI) measure to verify how attention affects imbalance in common stock buying/selling behavior.

Because we primarily focus on option contracts we adjust this common-stock based measure to an interpretable imbalance proxy for option contracts. One way to do so is by clustering ‘bullish’ option trades (long calls + short puts) and to subtract ‘bearish’ option trades (short call + long puts) and express this as a percentage of all option trades.¹² In order to avoid undue influence from any particular type of option contract to be bought or sold far out-of-the-money as opposed to other contracts, we delta-weight the option trades based upon the Black-Scholes option delta using 3-month EURIBOR rate as a proxy for the risk free rate.¹³ Algebraically it follows:

$$BBI_{pt} = \frac{\sum_{j=1}^{nQx} (|\delta_{LC_j}| + |\delta_{SP_j}|) - \sum_{j=1}^{mQx} (|\delta_{SC_j}| + |\delta_{LP_j}|)}{\sum_{j=1}^{nQx} (|\delta_{LC_j}| + |\delta_{SP_j}|) + \sum_{j=1}^{mQx} (|\delta_{SC_j}| + |\delta_{LP_j}|)} \quad (2)$$

BBI_{pt} refers to the Bull-Bear Imbalance measure; nQx is the number of *bullish* option contracts in partition p on day t ; mQx is the number of *bearish* option contracts in partition p on day t ; $\delta(\cdot)_{it}$ is the Black-Scholes option delta for option contract j on common stock i at day t , where a distinction is made between (1) long call contracts (LC), (2) short put contracts (SP), (3) short call contracts (SC), and (4) long put contracts (LP). We use 3-month EURIBOR rate as the proxy for the risk free rate and assume perfect foresight of the dividends paid by the underlying stock over the remaining life of the option. We then calculate the time-series mean of the daily imbalances for the days that we have trading data for the option traders.

¹² This method is related the net long ratio as developed in Mahani and Poteshman (2008) which aggregates open interest for a given stock across all strike prices and maturities [page 640].

¹³ Note that results are qualitatively very similar when applying equal-weighted imbalance calculations (See Appendix A.1).

These tests allow examinations of attention-driven trading imbalance patterns for individual investors' option trades. In the main analyses, we compare deciles imbalances for all investors with survey subsamples and split samples based upon information that relates to an investor optimism score.

3.3.2 Long-term Return Specification Tests

We next move beyond short-term tests and investigate the persistence of feedback strategies while controlling for important reference points as in Lakonishok et al. (2007) and Schmitz and Weber (2007). While short term impulses like prior day returns may be important triggers for individual investors' trading decisions (e.g., Barber and Odean 2008; Huddart et al. 2009), longer horizon feedback may be an equally plausible additional candidate that need particular attention. In particular, we test the importance of short- (< 1week), medium- (<1 month) and long-term returns (<1 year) as explanatory variables for individual option trading initiations. We estimate the following logit model:

$$\begin{aligned} \text{Option}_{\text{TRANS}}(1/0) = & \alpha + \beta_1.R_{\text{day}} + \beta_2.R_{\text{week}} + \beta_3.R_{\text{month}} \\ & + \beta_4.R_{6\text{month}} + \beta_5.R_{\text{year}} + \Phi_j. \text{CTRLS} + \varepsilon \quad (3) \end{aligned}$$

Where $\text{Option}_{\text{TRANS}}(1/0)$ is set equal to 1 for all investor-days on which a specific investor performs a long or short stock option trade and is equal to zero alternatively. Further, R refers to the compound return and the subscript stands for the measurement period: i.e., on the same day (R_{day}); from day -5 to -1 (R_{week}); from day -21 through -6 (R_{month}); from day -128 through -22 ($R_{6\text{month}}$), and from day -252 through -129 (R_{year}). CTRLS refers to a vector of control variables identified in prior literature as

determinants of investors' trading decisions (Huddart et al. 2009; Lakonishok et al. 2007) and include return volatility; modified moneyness; as well as other yardstick prices of the underlying common stock like 20 days high, low and averages. We interpret findings of these long-term return specification tests as additional evidence on the importance of short versus medium and long-term return feedback in option trading for the cross-section of individual investors and across optimism groups.

4. Results

4.1 Descriptive Statistics

Little is known about individual investors' option trading activities in general and about the database used in this study, so we provide relatively detailed descriptive statistics. Table 1 provides general statistics on the number of option trades we observe in the investor transaction sample over the time period 2006-2007. Of all 206,534 option transactions, the proportion of purchases is roughly equal to the proportion of sales (103,851 or 50.3% versus 102,683 or 49.7%). About 58% of all option transactions relate to calls contracts (119,118) and 42% relate to put contracts (87,346).

*****Insert Table 1 about here*****

About one third of all option trades refer to stock index (AEX) options. The remainder (138,366 transactions, or 67%) relate to individual common stocks. Long call stock option contracts occur more often (43,411 – or 31.4%) compared to short call option contracts (40,464 – or 29.2%). By contrast, individual investors write more put options (30,541 – or 22.1%) than they buy (23,950 – or 17.3%) over the time period of consideration. Proportions look fairly identical when we select only *naked* stock option

transaction, i.e. contracts for which the investor does not hold the underlying common stock at the same time.

These numbers indicate that the hedging motive (of which the combination of holding a stock with a long position in a put is the textbook example) is not the dominant reason for trading options. These numbers confirm earlier findings for US investors by Lakonishok et al. (2007) though that covered call options belong to the most popular option strategies (about 30%), but indicate that naked call writing is equally popular for Dutch private investors. Furthermore, we find that bullish transactions (buying calls and writing puts) slightly exceeded bearish transactions over the sample period (53.5 percent versus 46.5 percent). Finally, stock option transactions from survey respondents consist of 38,266 unique transactions, representing about 28% of total stock option transactions. The general observation of a higher proportion of long versus short calls and a lower proportion of long versus short puts is observed in all sub-samples except for index options.

*****Insert Table 2 about here*****

Table 2 provides more details on the actual transactions at the individual investor level, partitioned by option type. In Panel A, we provide statistics for the full sample whereas panel B contains statistics for the survey sample. The median investor initiates 18 option contracts during the study period of which 13 are done in stock options. Less than half of the investor trade in index options (median=0). Further, the 99th percentile indicates that some large traders account for a high number of option

trades.¹⁴ Also, we find that the median investor trades three times more frequently in calls options than in put options (calls: 12 versus puts: 4). This suggests that a larger fraction of all put options are initiated by a minority of investors. Black-Scholes option deltas, capturing the sensitivity of an option price to changes in the underlying asset's price, are below 0.50 for more than 75% of all observations and suggest that the majority of the call option contracts are out-of-the-money at initiation date. This is far less the case with put options (median=-0.62).

Further, we find that the median stock option contract is executed at a value of 821 EURO while one quarter of all options is traded at values above 2,000 EURO. Absolute values of call options are generally higher compared to put options (median long call=916 EURO compared to median long put=680 EURO). Finally, and not surprisingly, we find that the average transaction amount for common stock is substantially larger than for options. The median investor trades stock for an amount of 4,900 EURO. One quarter of the stock transactions are above 10,400 EURO.

Finally, results for the survey sample are very similar to results for the full sample. We only describe some essential statistics for the reason of brevity. Median option trades now equal 21 compared to 18 for the full sample. Again for the median investor, we observe somewhat more call option trades (15 versus 12) and about the same put option trades (3 versus 4). Absolute EURO amounts of option contracts are now more equal for call and put options (744 EURO for long calls and 739 for long puts). Finally, the average

¹⁴ We perform sensitivity checks were we exclude the top 2% and top 5% of extreme active investors and results remain qualitatively unchanged. See section 5.2 for more details.

EURO amount of common stock transactions for the surveyed sample is very similar to the one for the full sample.

*****Insert Table 3 about here*****

On the survey sample, we report additional personal characteristics in Table 3 and compute differences in sample medians for investors that trade in common stock as opposed to investors that trade in common stock and/or index options. We find that the median option investor is significantly more experienced (6 to 10 years of experience, compared to 4 to 6 years for common stock traders [$p < 0.001$]) and spends more time on his/her investment portfolio per week (up to 2 hours per week vis-à-vis less than 1 hour per week for a common stock investor [$p < 0.001$]). There is no observable difference in optimism scores across investor groups: the median investor in both sub-groups invest a fairly high proportion of the total capital available for investments (between 80% and 90%) at the retail bank of investigation [$p = 0.822$]. In terms of portfolio composition, we observe similar proportions of common stock and closed-end funds, although the proportion of derivatives is – by construction – lower for the stock only investor group.¹⁵

4.2 Buy-Sell Imbalances of Common Stock

Before moving to imbalance evaluations in stock option trading behavior, we first examine to what extent findings for US individual investors as in Barber and Odean (2008) are present in our sample for Dutch individual investors. We calculate BSI measures where we sort stocks daily into 10 deciles on the basis of the current day's

¹⁵ Note that 25% of stock-only investors report that derivatives are maximally 10% of the family investment portfolio; a finding that might seem contra-dictionary at first sight. An investor holding index options *only* or holding derivatives like futures or warrants but no stock options, may explain this proportion to be different from zero.

abnormal volume (Panel A: *Volume* sorted) and on the basis of prior day's return (Panel B: *Return* sorted). Column (1) reports results for all investors and Column (2) is for the survey sample only. As a comparison group, we summarize the numbers for small discount brokerage investors (Column 3) and value managers (Column 4) as reported in Barber and Odean (2008). We choose these two investor characteristics because the former should be most comparable to the set of investors we observe in terms of investment profile and the latter is a benchmark group for sophisticated investors.

*****Insert Table 4 about here*****

*****Insert Figure 1a and 1b about here*****

We learn from panel A of Table 4 that the buy-sell imbalance is -12.2% for stocks in the lowest volume decile ($t=-2.94$) and +12.3% for the highest volume decile ($t=4.51$).¹⁶ Results for survey investors are similar for the highest abnormal volume decile: +12.0% ($t=4.17$) but less pronounced for the decile with lowest abnormal volumes (-5.0% but insignificantly different from zero). When we compare these numbers in extreme volume deciles with US results (Column 3 and 4), we find that investor imbalances are rather similar to those of small retail bank investors (Column 3) but not to those of value managers.

Panel B of Table 4 as well as Figure 1b provide more details on buy-sell imbalances for deciles sorted on prior day's return. Interestingly, we observe that for both the full sample (Column 1) as the survey sample (Column 2), investors highly overweigh purchases compared to sales for poor prior day return performance stocks. The buy-sell imbalance is as high as 33.2% (i.e., about 66 percent of all trades are

¹⁶ t -statistics are based on Newey-West corrected standard errors, corrected for serial dependence.

purchases) for the full sample and even 41.6% (i.e., about 72% of all trades are purchases) for the survey sample. At the other end of the prior day return spectrum, we observe considerable underweighting of extremely good performing stock: -13.1% for all investors and even -23.6% for survey sample investors. Both observations suggest that individual investors follow a contrarian investment strategy where prior day's winners are sold and prior day's losers are bought. This is in line with the investment pattern of value managers (Column 4) and inconsistent with that of small discount brokerage traders (Column 3) in a US setting.¹⁷

Combined, this suggests that individual investors are net buyers of common stock on high attention days, when attention is measured by abnormally high contemporaneous trading volume and by extremely poor prior day stock returns. However, the persistent and opposite buy-sell imbalance for extremely positive return stock (i.e., making them net sellers of these stock) suggest that individual investors also pursue a short-term negative feedback (i.e., contrarian) investment strategy.¹⁸ In the next section, we explore whether and to what extent individual traders use stock option trades to further substantiate these investment styles.

4.3 Bull-Bear Imbalances of Stock Options

If investors, be it professional or individual investors, follow return feedback investment strategies, stock option trading would be a perfect mechanism to speculate

¹⁷ Interestingly, where Barber and Odean (2008) only observe contrarian investment preferences for the top-decile of positive return portfolios our results suggest that contrarian styles also occur as from decile 6 and onwards (see also Figure 1b).

¹⁸ A devil's advocate may attribute this finding to the fact that investors in our sample primarily will select out of 75 underlying domestic (Dutch) stocks and that every investor may hold a large majority of these stocks, and hence can sell in case of bad news. Unreported tests, however, show that the average (median) number of different underlying stock an investor holds is 8 (6) which is far below this total number of 75. This suggests that buy-sell imbalances for our sample are not just an artifact from the sample.

on market volatility and directional price movements (Hull 2008). We therefore calculate delta-weighted buy-sell imbalances for call and put option stock contracts and bull-bear imbalances as calculated by formula (2).¹⁹ In addition, we report delta-weighted call/put buy-sell imbalances to demonstrate that the bull-bear imbalances occur symmetrically across call and put options. For reasons of brevity, we expose on the bull-bear imbalances only, again for returns sorted (Panel A) and volume sorted (Panel B) deciles.

*****Insert Table 5 about here*****

*****Insert Figure 2 about here*****

Results show that abnormal volumes are not systematically related to the initiation of bullish versus bearish option contracts.²⁰ Consistent with the descriptive evidence in 4.1 above, we observe a rather stable pattern of slightly more bullish stock options (long calls + short puts) versus bearish stock options (short calls + long puts). However, when we sort deciles based upon prior day returns, individual investors initiate significantly more bullish-type option contracts for extremely poor performing stock (BBI= +42.8%, corresponding to almost 2.5 times as many bullish versus bearish stock options contracts). Likewise, individuals initiate significantly more bearish-type option contracts for extremely good performing stock (BBI = -21.6%, or corresponding to more than 50

¹⁹ We report delta-weighted imbalances because descriptive statistics (See Table 2) suggest a large majority of call options are taken out-of-the-money whereas this is less the case for put options. In additional robustness checks, we ignore differences in sensitivities to underlying stock movements and calculate equal-weighted transactions but find similar results (See section 5.2 and Appendix A.1 for a more detailed description).

²⁰ For the reason of brevity, we only report BBI for the full sample in table format. Figure 2 illustrates that BBI results are very similar for the full sample as for the survey sample.

percent more bearish compared to bullish stock options).²¹ In general, the current results confirm that option trades are initiated to pursue short term return feedback strategies (contrarian) rather than that they would emerge from attention-grabbing heuristics.

***** Insert Table 6 about here *****

We next examine to what extent investors use medium- and long-term return feedback to initiate stock option trades and to benefit from directional price movements. To that end, we run logistic regressions as described above in Section 3.3.2. Results are consistent with the short-return window BBI results in that very short term (i.e., same day) performance is negatively related to a bullish option initiation (long calls + short puts), after controlling for various other characteristics of the underlying. However, medium to long-term return feedback is always positively related to bullish option initiations, indicating that individuals try to make advance of unusual, one-off price dips when initiating bullish option transactions.

Results for bearish option trades are also consistent with short-run (1 day) negative and positive longer term return feedback trading. However, we find some disparity between long puts and short calls: investors buy put option contracts in case of extremely good short-term (1 day) performance which does not reflect medium term (< 1 month) performance, consistent with the idea that investors in long put options are following a contrarian strategy using medium-term negative feedback. Individuals write call options, by contrast, for stock that performed persistently well over both the short and long term (all coefficients on the past returns until $\text{Return}_{6\text{month}}$ until are significantly

²¹ Note that the decomposed call/put buy-sell imbalances suggest that whereas the bull-bear imbalance is driven equally by long calls and short puts, there are somewhat more short calls ($\text{BSI}_{\text{calls}} = -23.5\%$) compared to long put contracts ($\text{BSI}_{\text{puts}} = +19.9\%$) in case of extremely positive prior day performance.

positive). This suggests that investors not only use prior day extreme positive performance but also find confirmation in the longer time period price run-up of the underlying stock to motivate the decision to write a call contract.

These results further provide consistent evidence in line with hypothesis H1 in that in the absence of short sales constraints, individual investors use short as well as long-term return feedback to corroborate specific option trade decisions.

4.4 Behavioral Biases and Imbalances across Attention Portfolios

4.4.1 Optimism and Short-Term Feedback

We now turn towards the cross-sectional variation in option investment styles across investor optimism groups. In panel A of Table 7, we compare stock option BBIs across 5 quintiles that are sorted daily on the previous day's returns for extreme optimists (top quartile of investors [$N=241$]) versus extreme pessimists (bottom quarter of optimism score [$N=241$]). The top quartile of extreme optimists score 4.20 or higher on a maximum of 5.0 on the recoded LOT-R optimism score. Extreme pessimists score less than 3.50.

*****Insert Table 7 about here*****

We learn from Table 7 that contrarian investment actions in case of extreme negative returns (Quintile 1) are most likely to occur in the extreme optimists subsample (+0.450, or corresponding to about 2.6 times as many bullish compared to bearish option contracts). This is substantially higher than the BBI for the extreme pessimists subsample investors but only significant at the one-sided 10 percent level: (+0.081; $t=1.49$). In the case of extreme positive returns (Quintiles 4 and 5), we observe an opposite pattern and extreme optimists initiate significantly less bearish option transactions compared to

extreme pessimists (Q4: +0.106 [$t=2.17$], Q5: +0.119 [$t=2.40$]). This result is consistent with the idea that extreme optimists are the most optimistic at either sides of the return spectrum. In other words, the most optimistic investors – as opposed to least optimistic investors – are *more* convinced that extreme negative returns will revert in the short run and are at the same time *less* convinced that extreme positive returns will mean-revert in the short run. This finding is consistent with well-established findings in the psychology literature on trait optimism.

An interesting question then emerges whether extreme optimists are *always* more optimistic, and take more bullish stock option position independent of prior investment outcomes. One way to tackle this question is by slicing the sample into initial or opening transactions, i.e. the first option contract that we observe referring to a specific underlying, and compare these to subsequent transactions. A subsequent transaction differs from an initial transaction in that it refers to a follow-up option contract that follows the first (i.e., initial) option contract referring to the same underlying, but which can have different characteristics on strike price or contract type.²² Results are displayed in Section 2 of Panel A in Table 7 and suggest that differences between extreme optimists and pessimists for all option contracts are confirmed and are even more pronounced for initial (opening) option contracts. Extreme optimists are also here *more* convinced that extreme negative returns will revert in the short run (+0.144; $t=1.62$) and are at the same time *less* convinced that extreme positive returns will mean-revert in the short run (+0.312; $t=3.71$).

²² A potential flaw in this test is that we do not necessarily observe the investor's very first option contract ever on a specific underlying. To the extent that this relation is indeed affecting our results, it would rather work against finding results on a learning effect.

Results are especially intriguing when we look at the BBIs for extreme optimists in isolation. Whereas extreme pessimists are on average more bearish after extremely positive prior-day returns for initial option contracts (-0.139 ; $t=-2.89$), extreme optimists are more bullish on average for their initial option contract into stocks with similar characteristics ($+0.195$; $t=3.83$). Moreover, extreme optimists have a positive BBI over all 5 return quintiles suggesting extreme optimists are bullish-biased in their initial trades, independent of the prior day return signal. Results are very similar for initial *and* naked option contracts (i.e. the first option contracts on a specific underlying that we observe, provided that the investor does not hold the underlying). However, because of this two-step conditional selection, observations become very scarce and lead to power issues for significance testing. We find that our results are insensitive to shifting the optimism cut-off level to 20-60-20 or 15-70-15 percentiles although this decision leads to a reduction in observations in the extreme portfolios creation. The evidence is consistent with the argumentation of hypothesis H2 and shows that investor optimism largely contributes to an investor's preference set. In addition, it suggests that contrary to findings for gamblers (Gibson and Sanbonmatsu 2004), optimistic option investors do not necessarily maintain positive expectations in an environment where they can learn from prior actions.

4.4.2 Optimism and the Incremental Use of Long-Term Return Feedback

Panel B of Table 7 combines results of clustered bullish (long call + short put) and clustered bearish (long put + short call) option trades in a similar way as in Table 6 above, but now also for the sub-sample of survey respondents. We interact a dummy

variable (D_OPT) that is set equal to one if the investor is classified as an extreme optimistic investor with the return variables of interest to examine differences in return feedback usage across investor types.²³

We find evidence consistent with the full sample results in that investors use short- and long-term return feedback for initiating option trading decisions. However, results for bullish option trades also show that the coefficient on $D_OPT \times R_{day}$ is almost double the size of the coefficient on R_{day} (-13.22 versus -7.03) confirming that short-term return feedback usage is especially more prevalent in option trading decisions of extreme optimists compared to extreme pessimists, even after controlling for long-term return feedback and a variety of other determinants of investors' option trading decisions. Also, we observe that negative 1-week feedback is basically ignored by extreme optimists ($R_{week} + D_OPT \times R_{week} = +0.91$ with an insignificant t-value [*not reported*]) while it is part of the investment decision of less extreme optimists (-4.20; $t = -8.61$). The same findings are generally true for bearish option trades, although the coefficient size for $D_OPT \times R_{day}$ is lower in magnitude. Results are very similar (*not reported*, available upon request) for the *first* stock option contracts observed at the investor-underlying level (i.e., initial transactions). Combined, these results illustrate that extreme optimists react more positively on short-term return signals as opposed to less extreme optimists.

5. Robustness Checks

5.1 Equal-weighted Bull-Bear Imbalances

²³ Note that the reference group here includes all investors that do not belong to the top-quartile of most optimistic investors (i.e. extreme optimists). Results are similar if we exclude observations between Q1 and Q3 and take the bottom quartile of optimist scores (i.e., extreme pessimists) as reference group.

Our test metric for bull-bear imbalances implicitly includes sensitivities of option prices to underlying stock movements in aggregating option transactions by delta-weighting the option positions. The main motivation to do so is to control for potential biases in imbalance metrics in situations where buy and sell alike transactions are clustered in particular moneyness regions. At the same time, delta-weighting may be excessively cumbersome and potentially over-estimates imbalances in cases where particular investors take *extreme* out-of-the-money positions. We therefore also perform all our BBI tests with unweighted values and reach very similar conclusions. We insert Appendix A.1 as an illustration of how close *unweighted* BBI measures get to *delta-weighted* BBI metrics for return sorts and volume sorts deciles.

5.2 Naked versus Covered Option Positions

We performed sensitivity analyses on all stock option contracts for which the individual investor does not hold underlying stock in his/her portfolio and does not have other stock options positions (either long or short) on the same underlying. This analysis of subset of stock options should enable us to filter out the very true opinions by individuals on the underlying stock price movements without any type of bias from combined trading strategies. Imposing this selection criterion reduces the number of available transactions by more than half. We refer to Section 3 in Panel A of Table 7 to illustrate this reduction in available observations. Appendix A.2 summarizes the delta-weighted BBI results for naked option transactions. Despite the drop in usable observations, however, results for all tests are qualitatively similar to the ones we report in the main analyses of this study. We reside to the largest sample and are confident that

our results are not biased by unobserved option combinations such as covered calls and/or protective puts.

5.3 Initial versus Subsequent Stock Option Transactions

In accordance with sample splits that we report on investor optimism portfolios (Table 7), we also performed all other BBI analyses for the first (i.e., initial) stock option transactions we observe. These are stock option transactions for a particular underlying for which we do not observe prior investor-level activity (buy/sell) relating to the underlying common stock or to another stock option contract on the same underlying. For all tests, available observations drop when we restrict our sample to fit these criteria but results remain qualitatively unchanged. Appendix A.3 summarizes BBI results as reported in Table 5 for initial (i.e., opening) stock option positions and yields results very close to the ones observed for the full sample. This provides comfort that the general findings are not biased by some cumulative learning effects over the time period observed.

5.4 Other Robustness Tests

We perform a number of extra robustness checks to rule out alternative explanations for our findings and summarize the conclusions here. First, we exclude the 2%, respectively 5%, of most active traders (in terms of number of trades and self-declared weekly hours spent on investment portfolio) to control for the potential influential effect of these active traders on the observed relations. Investors which execute more than 250 stock option trades per year and/or spend more than 20 hours per week on their investment portfolio may have very different characteristics from a typical noise trader.

When we exclude observations from the most active investors, results continue to hold for all our tests, yielding confidence that the observed relations are generalizable to the average individual investor.

Second, all our tests implicitly assume that investors trade the large majority of their stock option contracts at the retail bank of investigation. To overcome this potential problem of not observing the true range of all stock option investments at the individual investor level, we exclude investors that have less than 75 percent of their investment capital traded via the retail bank of interest. About one third of the investors drop out of the sample and the number of available individual stock option contract observations reduces to about 20,000. Results for all tests remain qualitatively similar, however, after imposing this criterion.

Finally, we performed our tests above for a sample of index option contracts where data requirements allowed for incorporating index options in the analyses, i.e. where tests are not based upon abnormal volume specifications. Although the occurrence of index option transactions is substantially smaller (i.e., about half of stock option contracts), results generally yield qualitatively similar conclusions.

6. Conclusion

Recent anecdotal evidence suggests that especially in the aftermath of the recent financial crisis, individual investors become more and more attracted by the leveraged return features from trading in derivative instruments such as stock options (e.g., Collins and Kearns 2010). It is however a widespread belief that trading in options is “*one of the all-time suckers’ bets*” and that individual investors, being under-informed by nature,

cannot benefit from trading in such a complex type of derivative instruments. This invokes the logical question why individuals decide to take on option investment decisions in the first place; and even more so how personal trait characteristics like individual optimism relate to option investment decisions.

Puzzled by this research question, this paper uses a large dataset of over 200,000 option trades executed by over 2,200 individual investors to examine whether attention-type events dominate (long) return feedback strategies in explaining individual investors' stock option trading decisions. We show that individual investors especially exploit *stock options* to follow contrarian investment strategies in that they initiate over twice as many bullish-type (half as many bearish-type) option contracts after extremely poor (good) prior-day returns. This suggests that in the absence of short sales constraints, individual investors' investment behavior cannot be purely explained by naïve high-attention events. Moreover, we observe variance in contrarian behavior across investor types: extremely optimistic investors follow contrarian investment strategies more (less) pronounced around extreme negative (positive) prior day returns. Finally, we show that the same extreme optimists especially overweigh short-term compared to long-term return feedback information in making individual stock option trading decisions. Both findings are consistent with the intrinsic features of individuals' trait optimism. In addition, we are amongst the first to show that individual stock option investors update prior overly optimistic beliefs about underlying stock: initial stock option positions are on average more bullish-biased compared to subsequent trades and the bias is especially observed for extreme optimists.

Combined, this study provides novel insights in the dynamics of individual investors' option trading decisions and in the distinctive roles of cognitive biases underlying this process. The academic relevance of the current findings is substantial for various study fields like behavioral finance, psychology and economics in general. The findings are also important for policy makers and non-profit organizations in that they may want to inform investors about the dangers of cognitive biases like excessive optimism, potentially causing individuals to take on overly-risky investment decisions. Finally, these findings may be potentially helpful for brokerage firms and retail banks that help investors obtaining positive returns.

7. Reference List

- Barber, B. and T. Odean (1999). The courage of misguided convictions. *Financial Analysts Journal* 55(6): 41-55.
- Barber, B. and T. Odean (2000). Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors. *Journal of Finance*, 55(2): 773–806.
- Barber, B. and T. Odean (2002). Online investors: do the slow die first? *Review of Financial Studies* 15(2): 455-487.
- Barber, B. and T. Odean (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Review of Financial Studies* 21(2): 785-818.
- Bauer, R., Cosemans, R. and P. Eichholtz (2009). Option trading and individual investor performance. *Journal of Banking and Finance* 33(4): 731-746.
- Bell, T., Landsman, W., Miller, B. and S. Yeh (2002). The valuation implications of computer stock option accounting for profitable computer software firms. *The Accounting Review* 77(4): 971–996.
- Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. *Journal of Political Economy* 81: 637–654.
- Cao, M. and J. Wei (2008). Option market liquidity: Commonality and other characteristics. *Journal of Financial Markets* 13: 20-48.
- Collins, M. and J. Kearns (2010). Options traders' ranks swell as amateur investors embrace 'Iron Condors'. Available at: <http://www.bloomberg.com>
- De Bondt, W. (1998). A portrait of the individual investor. *European Economic Review* 42: 831-844.
- Dhar, R. and N. Zhu (2006). Up close and personal: Investor sophistication and the disposition effect. *Management Science* 52(5): 726-740.
- Felton, J., Gibson, B. and Sanbonmatsu D. (2004). Preference for Risk in Investing as a Function of Trait Optimism and Gender. *Journal of Behavioral Finance* 4(1), pp. 33-40.
- French, J. and J. Poterba (1991). Investor sophistication and international equity markets. *American Economic Review* 81(2): 222-226.

- Friedman L., Weinberg A., Webb J., Cooper H. and S. Bruce (1995). Skin cancer prevention and early detection intentions and behavior. *American Journal of Preventive Medicine* 11: 59-65.
- Gibson, B. and D. Sanbonmatsu (2004). Optimism, Pessimism, and Gambling: The downside of Optimism. *Personality and Social Psychology Bulletin* 30(2); 149-160.
- Graham, J., Harvey C., and M. Puri (2009). Managerial Attitudes and Corporate Actions. *Unpublished manuscript Duke University and NBER*.
- Griffin, J. M., Jeffrey H. Harris and S. Topaloglu (2003). Investor behavior over the rise and fall of Nasdaq. *Unpublished manuscript Yale University*.
- Grinblatt, M., and M. Keloharju (2001). What Makes Investors Trade? *Journal of Finance*, 56(2): 589–616.
- Heath, C., S. Huddart and M. Lang (1999). Psychological factors and stock option exercise. *Quarterly Journal of Economics* (May edition): 601-627.
- Huddart, S., Lang, M. and M. Yetman (2009). Volume and price patterns around a stock's 52 week highs and lows: Theory and evidence. *Management Science* 55(1): 16-31.
- Hull, J. (2008). Options, futures and other derivatives. Prentice Hall.
- Kindleberger, C. (2000). *Manias, Panics, and Crashes: A History of Financial Crises*, 4th edition, New York: Wiley.
- Lakonishok, J., Lee, I., Pearson, N. and A. Poteshman (2007). Option market activity. *Review of Financial Studies* 20: 813-857.
- Mahani, R. and A. Poteshman (2008). Overreaction to stock market news and misevaluation of stock prices by unsophisticated investors: Evidence from the option market. *Journal of Empirical Finance* 15: 635–655.
- Murray S. (2001). Japan Goes Dutch. *London Review of Books* XXIII. April 5, 2001.
- Odean, T., 1998, Are Investors Reluctant to Realize Their Losses? *Journal of Finance*, 53(5): 1775–1798.
- Odean, T., 1999, Do Investors Trade Too Much? *American Economic Review*, 89(5): 1279–1298.
- Poteshman, A. (2001). Underreaction, overreaction, and increasing misreaction to information in the options market. *Journal of Finance* 56(3) – June 2001: 851-876.

Poteshman, A. and V. Serbin (2003). Clearly irrational financial market behavior: Evidence from the early exercise of exchange traded stock options. *Journal of Finance* 58: 37-70.

Puri, M. and D. Robinson (2007). Optimism and economic choice, *Journal of Financial Economics* 86: 71-99.

Rajgopal, S. and T. Shevlin (2002). Empirical evidence on the relation between stock option compensation and risk taking. *Journal of Accounting and Economics* 33: 145-171.

Roll, R., Schwartz, E. and A. Subrahmanyam (2010). O/S: The relative trading activity in options and stock. *Journal of Financial Economics* 96: 1-17.

Scheier M., Carver C. and M. Bridges (1994). Distinguishing optimism from neuroticism (and trait anxiety, self-mastery, and self-esteem): a reevaluation of the life orientation test. *Journal of Personality and Social Psychology* 67(6): 1063-1078.

Scheier M., Carver, C. and M. Bridges (2001). *Optimism, pessimism, and psychological well-being*. In E. C. Chang (Ed.) *Optimism and pessimism: Implications for theory, research, and practice*. (189-216). Washington, DC: American Psychological Association.

Schmitz, P. and M. Weber (2007). Buying and Selling Behavior of Individual Investors in Option-like Securities. *Unpublished manuscript Mannheim University*.

Segerstrom, S. (2001). Optimism and attentional bias for negative and positive stimuli. *Personality and Social Psychology Bulletin* (27): 1334-1343.

Statman, M. (2004). The diversification puzzle. *Financial Analysts Journal* 60(4): 44-53

Van Oostvoorn, M. and L. Rijnhout (2006) *Beleggen met opties: de beste strategieën met het hoogste rendement* (Dutch only). Pearson Education Benelux.

8. Tables and Graphs

Table 1: Option Contract Details

	TOTAL		CALLS		PUTS	
ALL – purchases	103,851	50.3%	62,377	30.2%	41,474	20.1%
ALL – sales	102,683	49.7%	56,811	27.5%	45,872	22.2%
Sum	206,534	100.0%	119,188		87,346	
INDEX – purchases	36,490		18,966	27.8%	17,524	25.7%
INDEX – sales	31,678		16,347	24.0%	15,331	22.5%
Sum	68,168	33.0%	35,313		32,855	
STOCK – purchases	67,361		43,411	31.4%	23,950	17.3%
STOCK – sales	71,005		40,464	29.2%	30,541	22.1%
Sum	138,366	67.0%	83,875		54,491	
NAKED – purchases	43,498		28,406	32.4%	15,089	17.2%
NAKED – sales	44,070		25,404	29.0%	18,663	21.3%
Sum	87,568	42.4%	53,810		33,752	
SURVEY – purchases	13,229		8,937	31.9%	4,292	15.3%
SURVEY – sales	14,763		8,152	29.1%	6,611	23.6%
Sum	27,992	13.6%	17,089		10,903	

This table summarizes the total number of stock and index option contracts for our sample of individual investors (N=2,254) over the period January 2006 – December 2007. INDEX refers to the Dutch AEX30-index; STOCK refers to common stock options; NAKED refers to the option contracts for which the investor does not the underlying stock; and SURVEY refers to the sample of investors on which we have psychometric information (N=399).

Table 2: Investor-Option StatisticsPanel A: Full Sample (N=2,254)

	N	P1	Q1	Median	Q3	P99	Std
<u>Number of trades</u>							
#option trades	206,534	1	5	18	70	1,292	287.3
#stock option trades	138,366	0	3	13	45	751	235.9
#index option trades	68,168	0	0	0	6	534	145.4
#calls	119,188	0	3	12	42	745	159.3
#calls long	62,377	0	1	6	21	415	81.8
#calls short	56,811	0	2	6	20	323	79.4
#puts	87,346	0	0	4	23	602	152.3
#puts long	41,474	0	0	1	10	296	74.1
#puts short	45,872	0	0	2	11	313	81.1
<u>Black-Scholes Option Delta</u>							
Calls	83,875						
Delta(calls long)	43,411	0.00	0.02	0.23	0.43	1.00	0.32
Delta(calls short)	40,464	0.00	0.02	0.24	0.44	1.00	0.33
Puts	54,491						
Delta(puts long)	23,950	-1.00	-0.83	-0.62	-0.16	0.00	0.41
Delta(puts short)	30,541	-1.00	-0.77	-0.63	-0.28	0.00	0.38
<u>Common stock Investments</u>							
Stock Option trade (Euro amount)	206,534	25	339	821	2,049	21,290	5,610
Long Call	62,377	30	383	916	2,198	19,002	5,269
Short Call	56,811	35	371	901	2,286	22,327	5,901
Long Put	41,474	10	249	680	1,805	23,001	5,795
Short Put	45,872	41	329	739	1,748	22,511	5,511
Stock trades (Euro)	195,482	160	2,271	4,900	10,451	109,002	29245.73

Panel B: Survey Sample (N=373)

	N	P1	Q1	Median	Q3	P99	Std
<u>Number of trades</u>							
#option trades	38,266	1	6	21	76	1,319	233.3
#stock option trades	28,001	0	4	16	52	759	179.4
#index option trades	10,265	0	0	0	4	411	109.9
#calls	22,155	0	4	15	44	801	129.3
#calls long	11,597	0	2	7	20	386	70.0
#calls short	10,558	0	2	7	20	340	60.6
#puts	16,111	0	0	3	26	779	124.4
#puts long	6,989	0	0	2	11	308	48.4
#puts short	9,122	0	0	2	14	356	83.3
<u>Black-Scholes Option Delta</u>							
Calls	17,089						
Delta(calls long)	8,937	0.00	0.05	0.25	0.45	1.00	0.32
Delta(calls short)	8,152	0.00	0.03	0.24	0.44	1.00	0.33
Puts	10,903						
Delta(puts long)	4,292	-1.00	-0.79	-0.56	-0.14	0.00	0.40
Delta(puts short)	6,611	-1.00	-0.69	-0.52	-0.16	0.00	0.38
<u>Common stock Investments</u>							
Stock Option trade (Euro amount)	38,266	22	304	736	1,771	20,303	4,782
Long Call	11,597	22	324	744	1,777	15,785	4,058
Short Call	10,558	29	309	752	1,914	20,812	5,105
Long Put	6,989	15	243	739	1,882	26,286	5,453
Short Put	9,122	30	316	716	1,553	21,329	4,687
Stock trades (Euro)	40,605	162	2,158	4,540	10,055	56,321	17,682

This table summarizes stock and index option trading by *individual* investors (N=2,254) over the period January 2006 – December 2007. Panel A contains observations for the full sample whereas panel B: Survey sample refers to the sample of investors on which we have psychometric information (N=373). Delta(.) refers to the Black-Scholes option delta using 3-month EURIBOR rate as the proxy for the risk free rate and assuming perfect foresight of the dividends paid by the underlying stock over the remaining life of the option.

Table 3: Investor Survey Characteristics (Option Traders versus Other Traders)

Variable	Type	N	Q1	Median	Q3
EXPERIENCE (in years)	Stock Only	612	4 to 6	6 to 10	>15
	Stock + Options	373	6 to 10	10 to 15 [<.001]	>15
TIMESPENT (in hours)	Stock Only	612	<1	1 to 2	4 to 6
	Stock + Options	373	1 to 2	2 to 4 [<.001]	6 to 8
OPTIMISM (R-LOT)	Stock Only	612	3.50	3.83	4.17
	Stock + Options	373	3.50	3.83 [0.880]	4.17
%WEALTH	Stock Only	612	40 to 50	80 to 90	100
	Stock + Options	373	50 to 60	80 to 90 [0.822]	100
%STOCK/%FUNDS/%DERIVATIVES	Stock Only	612	10/10/00	20/20/00	40/40/10
	Stock + Options	373	20/00/10	30/20/20	50/30/20

This table reports summary information for surveyed investors (N=985) of which 373 investors (37.9%) invest into individual stock options and/or common stock whereas 612 investors (62.1%) do not invest into stock options yet do invest into individual common stock over the period of observation (January 2006 – December 2007). *EXPERIENCE* refers to number of years that a respondent has active experience in trading common stock and/or derivative products. This question ranges from 1 (less than 6 months experience) to 8 (15 years or more of experience). *TIMESPENT* measures the average amount an investor spends on a weekly basis on his/her investment portfolio. Answers range between 1 (less than 1 hour per week) to 8 (20 hours or more per week). *OPTIMISM* refers to the recoded R-LOT score that ranges between 1 (very pessimistic) and 5 (very optimistic) and serves as a proxy for individual investors' optimism (Scheier et al. 1994; Graham et al. 2010). *%WEALTH* is the proportion of family wealth available for investment purposes that every specific investor reports to be investing at the retail bank of investigation; this measure represents the inverse likelihood of trades potentially executed (and hence *unobserved*) at other retail banks. The variable *%STOCK/%FUNDS/%DERIVATIVES* reports the percentage of family wealth available for investment purposes every respondent invests into (1) individual common stock / (2) closed-end investments funds / (3) derivatives [among which: futures, warrants, options and speeders]. *P*-values are reported between square brackets under the medians and are obtained via a two-sided Mann-Whitney U test.

Table 4: Attention-Grabbing Imbalance Results

Panel A – Buy/Sell Imbalances Common Stock, Abnormal Volume Sorts

	All Investors		Survey Sample		Barber/Odean - Small Discount Brokerage		Barber/Odean - Value Managers	
Decile	BSI	t-stat	BSI	t-stat	BSI	t-stat	BSI	t-stat
1	-0.122	-2.94	-0.050	-1.09	-0.205	-0.17	0.346	6.24
2	-0.070	-1.93	-0.008	-0.22	-0.103	-4.48	0.152	6.47
3	0.008	0.23	-0.003	-0.08	-0.070	-4.73	0.110	7.35
4	-0.019	-0.61	0.019	0.48	-0.049	-4.21	0.100	8.15
5	-0.010	-0.33	-0.026	-0.65	-0.041	-5.27	0.109	9.91
6	0.024	0.78	0.070	1.96	-0.019	-2.30	0.087	8.48
7	0.071	2.53	0.069	2.06	-0.001	-0.07	0.073	7.47
8	0.040	1.52	0.082	2.23	0.014	1.81	0.089	9.40
9	0.129	4.94	0.086	2.73	0.058	9.32	0.078	7.75
10	0.123	4.51	0.120	4.17	0.131	14.19	0.063	3.99
#Traders	3,011		564		n.a.		n.a.	
#Trades	100,665		23,603		n.a.		n.a.	

Panel B – Buy/Sell Imbalances Common Stock, Prior Day's Return Sort

All Investors			Survey Sample		Barber/Odean - Small Discount Brokerage		Barber/Odean - Value Managers	
Decile	BSI	t-stat	BSI	t-stat	BSI	t-stat	BSI	t-stat
1	0.332	11.65	0.416	14.92	0.157	13.63	0.156	5.77
2	0.222	7.66	0.276	8.65	0.055	5.47	0.112	8.81
3	0.182	5.85	0.127	3.36	0.038	4.96	0.102	9.65
4	0.010	0.32	0.045	1.09	0.017	2.01	0.080	8.06
5	0.065	2.03	0.078	2.15	-0.006	-0.67	0.092	7.13
6	-0.041	-1.14	-0.055	-1.34	-0.010	-1.21	0.090	4.99
7	-0.034	-1.09	-0.067	-1.81	-0.018	-2.16	0.106	8.99
8	-0.036	-1.15	-0.164	-4.65	-0.015	-1.87	0.079	7.47
9	-0.128	-4.11	-0.121	-3.41	0.006	0.75	0.043	3.55
10	-0.131	-4.26	-0.236	-8.84	0.025	2.87	-0.107	-3.73
#Traders	3,011		564		n.a.		n.a.	
#Trades	100,665		23,603		n.a.		n.a.	

This table reports buy-sell imbalances for common stock trades where stocks are sorted daily into deciles of the basis of the abnormal trading volume [Panel A] and previous days' return [Panel B]. Decile 1 (D1) refers to extremely low abnormal volumes, respectively extremely bad performance; and decile10 (D10) refers to extremely high abnormal volume, respectively extremely good performance. We calculate number imbalance as number of purchases minus number of sales divided by total number of trades (more information in 3.2 Research design). The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, and corresponding *t*-statistics are calculated using a Newey-West correction for serial dependence.

Table 5: Delta-weighted Bull-Bear Imbalance Results

Return sorts						
Decile	BBI	t-stat	BSI _{call}	t-stat	BSI _{put}	t-stat
1	0.428	22.01	0.381	13.07	-0.451	-18.22
2	0.321	14.39	0.266	8.74	-0.365	-13.17
3	0.241	11.07	0.185	6.24	-0.301	-11.34
4	0.186	7.61	0.103	3.04	-0.238	-8.27
5	0.132	6.11	0.053	1.63	-0.186	-7.04
6	0.094	4.05	0.024	0.78	-0.169	-5.71
7	0.010	0.46	-0.041	-1.26	-0.071	-2.69
8	-0.017	-0.76	-0.078	-2.41	-0.038	-1.45
9	-0.107	-4.63	-0.115	-3.62	0.091	3.19
10	-0.216	-10.48	-0.235	-8.58	0.199	7.77

Volume sorts						
Decile	BBI	t-stat	BSI _{call}	t-stat	BSI _{put}	t-stat
1	0.095	2.80	0.051	1.19	-0.207	-5.03
2	0.127	4.48	0.097	2.79	-0.157	-4.55
3	0.091	3.59	0.035	0.99	-0.124	-4.33
4	0.094	3.87	0.019	0.60	-0.126	-4.23
5	0.111	4.70	-0.003	-0.09	-0.167	-5.92
6	0.091	4.17	-0.014	-0.42	-0.148	-5.86
7	0.105	4.76	0.063	2.00	-0.135	-5.17
8	0.124	5.76	0.094	3.33	-0.158	-5.85
9	0.089	3.92	0.085	2.85	-0.116	-4.50
10	0.101	5.18	0.089	3.22	-0.107	-4.38

#Traders	2,127		2,127		2,127	
#Trades	138,366		138,366		138,366	

This table reports Black-Scholes delta adjusted bull-bear imbalances $f(BBI)$ for stock option trades where we define BBI as the weighted sum of the absolute deltas of bullish contracts [=A] minus the weighted sum of the absolute deltas of bearish contracts [=B] divided by the sum of [A] and [B]; (more information in 3.2 Research design). BBIs are further decomposed into BSI_{call} and BSI_{put} respectively, where the BSI measure also is delta-adjusted as discussed above. We sort the underlying common stocks daily into deciles of the basis of the previous days' return [Panel A] and abnormal trading volume [Panel B]. Decile 1 (D1) refers to extremely bad performance, respectively low abnormal volumes; and decile10 (D10) refers to extremely good performance, respectively extremely high abnormal volume. Standard errors, and corresponding t -statistics are calculated using a Newey-West correction for serial dependence.

Table 6: Short versus Long Term Returns and Individual Traders' Option Investment Decisions

	Bullish Trades		Bearish Trades	
	[1] Long Calls	[2] Short Puts	[3] Long Puts	[4] Short Calls
Constant	-3.944 *** (0.013)	-4.232 *** (0.015)	-4.801 *** (0.019)	-4.246 *** (0.015)
R _{day}	-4.165 *** (0.294)	-6.018 *** (0.358)	6.563 *** (0.392)	11.252 *** (0.320)
R _{week}	0.902 *** (0.148)	0.660 *** (0.182)	-2.001 *** (0.205)	0.715 *** (0.173)
R _{month}	1.495 *** (0.078)	0.950 *** (0.095)	-0.312 *** (0.114)	0.854 *** (0.084)
R _{6 month}	0.726 *** (0.029)	0.45 *** (0.036)	0.171 *** (0.042)	0.452 *** (0.031)
R _{year}	0.091 *** (0.031)	0.000 (0.001)	0.045 (0.043)	0.014 (0.032)
Volatility	18.581 *** (0.466)	15.581 *** (0.580)	15.728 *** (0.682)	6.297 *** (0.610)
Moneyness	0.01 *** (0.001)	0.013 *** (0.001)	0.016 *** (0.001)	0.011 *** (0.001)
Near_right	1.047 *** (0.036)	1.330 *** (0.037)	0.641 *** (0.069)	0.726 *** (0.056)
Near_wrong	1.165 *** (0.036)	0.754 *** (0.054)	0.808 *** (0.049)	1.11 *** (0.031)
Cross_right	1.421 *** (0.031)	1.538 *** (0.034)	0.816 *** (0.064)	1.103 *** (0.048)
Cross_wrong	1.234 *** (0.035)	1.130 *** (0.046)	1.118 *** (0.043)	1.32 *** (0.029)
D_max_19	0.491 *** (0.017)	(0.309 *** 0.021)	0.711 *** (0.021)	0.725 *** (0.015)
D_min_19	0.552 *** (0.018)	(0.461 *** 0.021)	0.532 *** (0.028)	0.073 *** (0.028)
D_avg	-0.096 *** (0.015)	-0.114 *** (0.018)	0.243 *** (0.022)	0.37 *** (0.017)
Number of observations	1,300,224	1,300,233	1,300,233	1,300,224
Pseudo R2	0.029	0.026	0.027	0.050

This table reports coefficients (standard errors between brackets) of running logistic regressions of purchased call and written put (i.e., bullish option transactions) versus purchased put and written call (i.e., bearish option transactions) on a number of explanatory variables. The explanatory variables include returns of underlying stocks with various cut-off dates: the same day return (R_{day}), the return from trade dates -5 through -1 (R_{week}), from trade dates -21 through -6 (R_{month}), from trade dates -128 through -22 (R_{6 months}), and from trade dates -252 through -129 (R_{year}). Control variables include the volatility of the underlying stock over the past 60 trading days (*Volatility*); moneyness of the underlying stock (*Moneyness*) measured

as $\frac{K + (S_t - K)}{K}$ for call options and $\frac{K + (K - S_t)}{K}$ for put options where K refers to the strike price of

the option and S_t is the price of the underlying asset; and a number of dummy variables that relate to yardstick prices of the underlying common stock. *Near_right* equals one if $0.98 < \text{moneyness}_{(t)} < 1$ and $\text{moneyness}_{(t-1)} < 0.98$ and zero alternatively; *Near_wrong* equals one if $1.00 < \text{moneyness}_{(t)} < 1.02$ and $\text{moneyness}_{(t-1)} > 1.02$; *Cross_right* equals one if $\text{moneyness}_{(t)} < 1.02$ and $\text{moneyness}_{(t-1)} < 1.00$ and zero alternatively; *Cross_wrong* equals one if $\text{moneyness}_{(t)} < 1.00$ and $\text{moneyness}_{(t-1)} > 1.00$ and zero alternatively. Other dummies include $D_{\text{max_19}}$ ($D_{\text{min_19}}$) and are equal to one if the underlying stock trades below the maximum (above the minimum) price over the past 19 trading days and zero alternatively. Finally D_{avg} is equal to one if the underlying stock trades above the average price of the past 19 trading days, zero alternatively. *, **, *** refers to significance at the 10%, respectively 5%, respectively 1% confidence level.

Table 7: Short versus Long-Term Return Feedback across Optimism Portfolios

Note: *Panel A* of this table reports bull-bear imbalances (BBI) for stock option trades where we define BBI as the weighted sum of the absolute deltas of bullish contracts [=A] minus the weighted sum of the absolute deltas of bearish contracts [=B] divided by the sum of [A] and [B]; (more information in 3.2 Research design). We cluster observations by optimism score for survey respondents where Pessimists are investors that score below 3.5 on the recoded R-LOT optimism score (bottom quartile) and extreme optimists score 4.2 or higher (top quartile), and where the theoretically maximum score equals 5. Quintile 1 (Q1) refers to extremely bad prior day performance; and quintile 5 (Q5) refers to extremely good prior day performance. The outmost right column contains bull-bear imbalance differences between portfolios (Optimists – Pessimists). Note that average differences in BBI percentages [(2) – (1)] are not equal to the arithmetic means of the optimist versus pessimists columns due to imperfect overlap of available BBI scores on specific days for each investor group. In all tests, standard errors, and corresponding *t*-statistics are calculated using a Newey-West correction for serial dependence. Panel A contains statistics for all common stock option transactions [1]; for initial stock option transactions at the investor-underlying level over the observed time period [2]; and for initial stock option transactions at the investor-underlying level where investors do not hold an investment in the underlying (i.e., for naked transactions) [3]. *Panel B* of this table reports coefficients (standard errors) of logistic regressions for bullish option transactions (i.e., purchased call and written put transactions) versus bearish option transactions (i.e., purchased puts and written calls) on returns of the underlying common stock measured across different time periods. The explanatory variables include the same day return (R_{day}), the return from trade day –5 through –1 (R_{week}), from day –21 through –6 (R_{month}), –from day –128 through –22 ($R_{6\ months}$), and from day –252 through –129 (R_{year}) as well as control variables as in Table 5 [not reported for reasons of brevity]. CTRLS is a vector of control variables described in Table 5. While panel A allows for identification of very short run contrarian/momentum investment strategies, panel B quantifies contrarian/momentum strategies for longer time periods in one regression framework. Note that the total number of trades (ALL) is larger in panel A compared to panel B due to data requirements for calculating longer-term return windows in panel B. *, **, *** refers to significance at the 10%, respectively 5%, respectively 1% confidence level.

Panel A: Delta-weighted Bull – Bear Imbalances, Prior day return sorts

[1] ALL

Quintile	(1) Pessimists (bottom Q)		(2) Optimists (top Q)		Difference [(2) - (1)]		
Q1	0.264	9.40	0.450	10.71	0.081	1.49	*
Q2	0.135	5.45	0.176	5.74	0.031	0.58	
Q3	0.059	1.49	0.032	3.22	0.051	0.86	
Q4	-0.147	-2.85	-0.095	-0.35	0.106	2.17	***
Q5	-0.288	-8.66	-0.180	-4.42	0.119	2.40	***
#Trades	10,134		9,046				

[2] INITIAL

Quintile	(1) Pessimists (bottom Q)		(2) Optimists (top Q)		Difference [(2) - (1)]		
Q1	0.359	7.41	0.496	11.00	0.144	1.62	*
Q2	0.322	5.91	0.354	6.99	0.006	0.08	
Q3	0.177	3.16	0.358	7.95	0.149	1.85	**
Q4	0.096	1.71	0.328	6.01	0.224	2.66	***
Q5	-0.139	-2.89	0.195	3.83	0.312	3.71	***
#Trades	2,869		2,792				

**[3] INITIAL
+ NAKED**

Quintile	(1) Pessimists (bottom Q)		(2) Optimists (top Q)		Difference [(2) - (1)]	
Q1	0.406	6.71	0.543	10.83	0.062	0.48
Q2	0.377	5.56	0.413	6.96	-0.090	-0.74
Q3	0.267	4.13	0.430	7.96	0.104	0.98
Q4	0.193	2.82	0.395	5.94	0.113	1.03
Q5	-0.117	-1.99	0.270	4.13	0.359	2.83 ***
#Trades	1,742		1,758			

Panel B: Short and Long-term Contrarian Option Investment Decisions
(Optimists versus Pessimists)

	Bullish trades (LC + SP)		Bearish trades (LP + SC)	
Constant	-4.896 *** (0.021)		-4.984 *** (0.022)	
R _{day}	-7.034 *** (1.013)		17.062 *** (0.776)	
R _{week}	-4.203 *** (0.488)		3.300 *** (0.520)	
R _{month}	-0.301 (0.264)		1.506 *** (0.277)	
R _{6 month}	0.226 ** (0.102)		-0.060 (0.112)	
R _{year}	-0.242 ** (0.112)		-0.248 ** (0.117)	
D_OPT	-0.076 ** (0.030)		-0.179 *** (0.033)	
D_OPTx R _{day}	-6.192 *** (1.334)		3.478 ** (1.085)	
D_OPTx R _{week}	5.112 *** (0.711)		-0.971 (0.731)	
D_OPTx R _{month}	0.552 (0.374)		-0.332 (0.387)	
D_OPTx R _{6month}	0.177 (0.143)		0.213 (0.154)	
D_OPTx R _{year}	-0.126 (0.157)		-0.146 (0.161)	
CTRLS	included		included	
Number of observations	1,148,251		1,148,251	
Nr. Of Ties	8,314		7,559	
Pseudo R ²	0.4%		1.1%	

Figure 1a: Common stock – Volume sorts

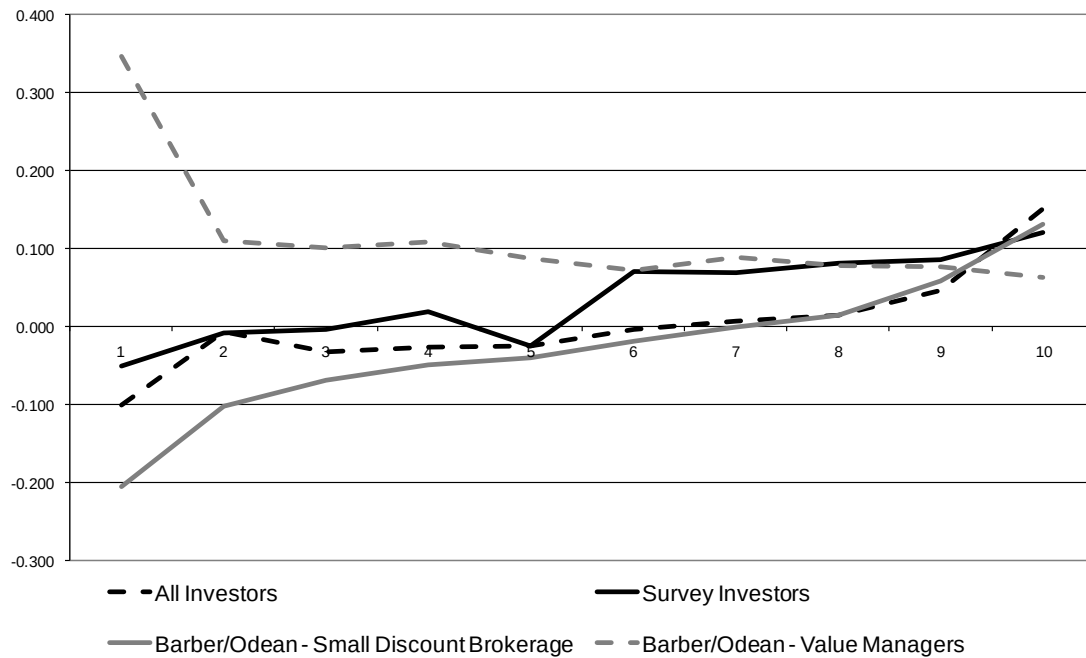
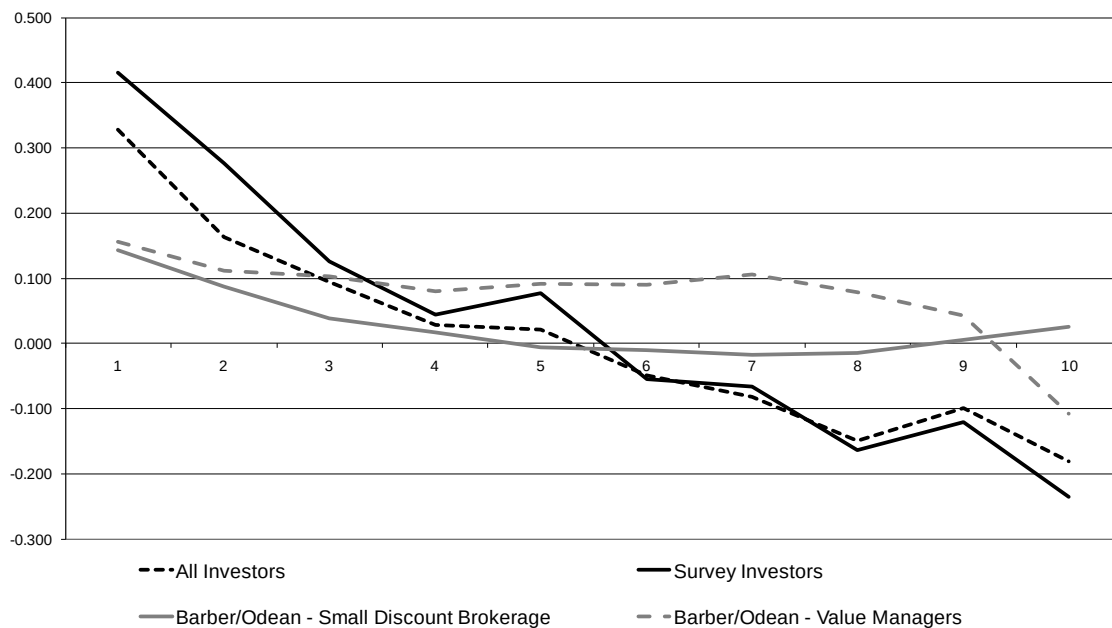
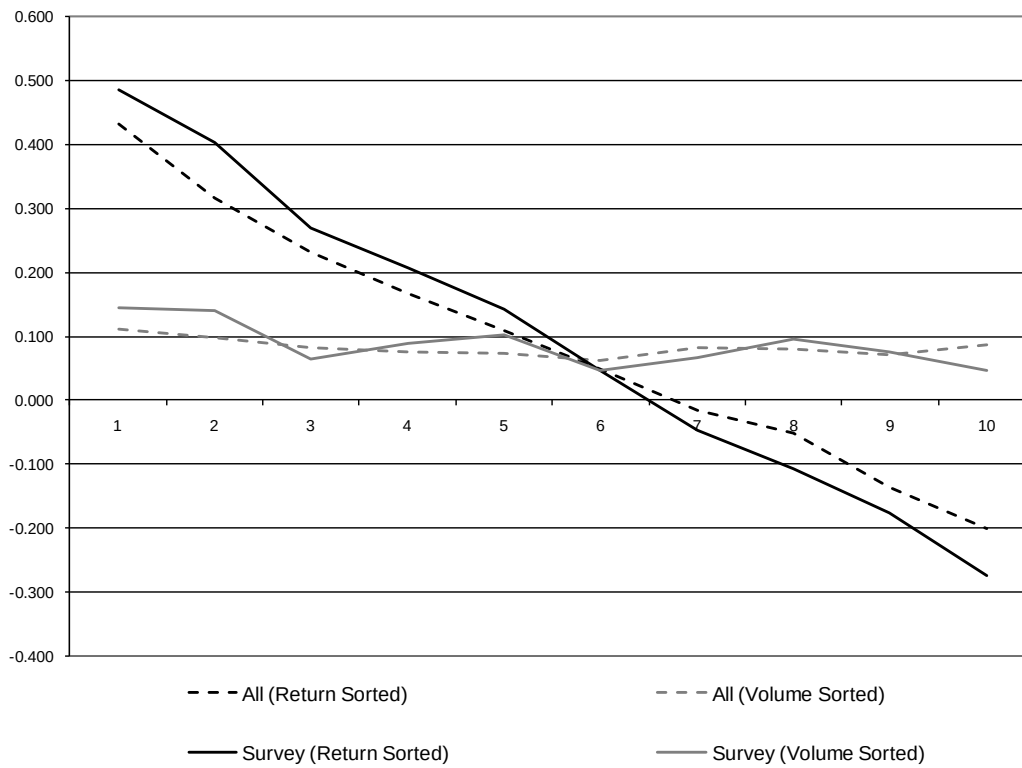


Figure 1b: Common stock – Return sorts



This figure reports buy-sell imbalance scores across 10 abnormal volume deciles (panel A), respectively prior day's return deciles (panel B) for all investors (dotted black line); survey investors (solid black line) as well as benchmark samples are reported in Barber and Odean (2008). The solid grey line corresponds to small discount broker observations and the dotted grey line refers to value manager observations, both as reported in Barber and Odean (2008).

Figure 2: Bull-bear Imbalances across Volume and Return-sorted deciles



This figure reports bull-bear imbalance scores across 10 abnormal volume deciles, respectively prior day's return deciles. *All* refers to imbalances for all investors; *Survey* refers to imbalances for survey respondents.

9. Appendices

Table A.1: Unweighted Bull-Bear and Buy-Sell Imbalance Results

Return sorts						
Decile	BBI	t-stat	BSI _{call}	t-stat	BSI _{put}	t-stat
1	0.439	27.79	0.428	21.84	-0.459	-21.44
2	0.316	18.66	0.287	13.56	-0.368	-15.22
3	0.225	13.26	0.201	9.22	-0.291	-12.32
4	0.174	8.65	0.121	4.72	-0.258	-9.84
5	0.106	6.22	0.061	2.55	-0.183	-7.60
6	0.060	3.33	0.020	0.90	-0.175	-6.50
7	-0.008	-0.45	-0.040	-1.62	-0.090	-3.70
8	-0.060	-3.25	-0.106	-4.77	-0.046	-1.80
9	-0.126	-7.10	-0.154	-7.02	0.066	2.56
10	-0.210	-12.91	-0.230	-12.10	0.149	6.30

Volume sorts						
Decile	BBI	t-stat	BSI _{call}	t-stat	BSI _{put}	t-stat
1	0.120	4.71	0.034	1.10	-0.218	-5.81
2	0.104	5.41	0.070	2.71	-0.175	-5.73
3	0.060	3.04	0.024	0.95	-0.137	-5.43
4	0.067	3.74	0.045	1.86	-0.134	-4.71
5	0.074	3.94	0.022	0.94	-0.173	-6.78
6	0.053	3.02	0.011	0.51	-0.155	-6.73
7	0.084	4.63	0.057	2.62	-0.149	-6.29
8	0.083	4.66	0.059	2.66	-0.155	-6.21
9	0.083	4.58	0.058	2.67	-0.142	-6.25
10	0.100	5.95	0.093	4.74	-0.142	-6.70

#Traders	2,127		2,127		2,127	
#Trades	138,366		138,366		138,366	

This table reports unweighted (i.e., not adjusted for Black-Scholes option delta weighting) bull-bear imbalances for stock option trades. More specifically, we define bull-bear imbalance as the number of bullish contracts minus the number of bearish contracts divided by the total number of stock options traded. Algebraically:

$$BBI_{pt} = \frac{\sum_{i=1}^{nQx} (N.LC_{it} + N.SP_{it}) - \sum_{j=1}^{mQx} (N.SC_{it} + N.LP_{it})}{\sum_{i=1}^{nQx} (N.LC_{it} + N.SP_{it}) + \sum_{j=1}^{mQx} (N.SC_{it} + N.LP_{it})} \quad (\text{App. F1})$$

BBI_{pt} refers to the unweighted Bull-Bear Imbalance measure; nQx is the number of bullish option contracts in partition p on day t ; mQx is the number of bearish option contracts in partition p on day t ; $N.LC_{it}$ is the number of long call contracts on stock i on day t ; $N.SP_{it}$ the number of short put contracts on stock i on day t ; $N.SC_{it}$ the number of short call contracts on stock i on day t ; and $N.LP_{it}$ the number of long put contracts on stock i on day t . We then calculate the time-series mean of the daily imbalances for the days that we have trading data for the option traders.. BBI are further decomposed into BSI_{call} and BSI_{put} respectively. We sort the underlying common stocks daily into deciles of the basis of the previous days' return [Panel A] and abnormal trading volume [Panel B]. Decile 1 (D1) refers to extremely bad performance, respectively low abnormal volumes; and decile10 (D10) refers to extremely good performance, respectively extremely high abnormal volume. Standard errors, and corresponding t -statistics are calculated using a Newey-West correction for serial dependence.

Table A.2: Delta-Weighted Bull-Bear Imbalance Results for *Naked* Option Positions

Decile	RETURN SORTS		VOLUME SORTS	
	BBI	t-stat	BBI	t-stat
1	0.457	21.59	0.148	1.44
2	0.315	13.44	0.137	3.31
3	0.231	8.73	0.082	2.32
4	0.185	5.34	0.083	3.24
5	0.112	2.53	0.090	1.86
6	0.072	2.09	0.094	2.34
7	0.027	0.09	0.103	3.04
8	-0.038	-2.33	0.086	2.75
9	-0.110	-4.94	0.094	3.00
10	-0.185	-9.00	0.120	6.24
# Traders	1,998		1,998	
# Trades	38,721		38,721	

This table reports delta-weighted bull-bear imbalances for naked stock option trades; i.e. for stock option trades for which the investor does not the underlying common stock. Full definitions on BBI calculations and attention-based sorts are described in-depth in Section 3.2.

Table A.3: Delta-Weighted Bull-Bear Imbalance Results for *Initial* Stock Option Transactions

Decile	RETURN SORTS		VOLUME SORTS	
	BBI	t-stat	BBI	t-stat
1	0.556	29.84	0.270	8.69
2	0.430	20.12	0.263	10.12
3	0.355	15.63	0.227	9.28
4	0.308	12.41	0.259	10.51
5	0.244	11.14	0.251	10.20
6	0.273	11.70	0.266	12.55
7	0.190	7.24	0.296	13.27
8	0.201	9.40	0.287	12.39
9	0.135	5.54	0.316	13.79
10	0.112	4.67	0.356	19.63
# Traders	2,127		2,127	
# Trades	56,961		56,961	

This table reports delta-weighted bull-bear imbalances for *initial* common stock option transactions; i.e. we solely include the first stock option transaction of individual investors for all underlying common stock on which investors enter into an option contract during the observation period (January 2006 to December 2007). Full definitions on BBI calculations and attention-based sorts are described in-depth in Section 3.2.